

Radar Signal Monitoring TO BE: 0.7.6.4.18.2.2

Review design for a planned Radar search pipeline change

Status: Implemented

Pipeline id: signal-monitoring

Baseline AS IS: docs/radar/pipelines/signal-monitoring/RADAR_SIGNAL_MONITORING_AS_IS.md

Baseline RCA: docs/radar/pipelines/signal-monitoring/diagnostics/SIGNAL_RUN_9d018757_863de7ce_QUALITY_RCA.md

Acceptance manifest: RADAR_SIGNAL_MONITORING_TO_BE_0.7.6.4.18.2.2.acceptance.json

Generated PDF: docs/radar/pipelines/signal-monitoring/to-be/RADAR_SIGNAL_MONITORING_TO_BE_0.7.6.4.18.2.2.pdf

1. Decision

Signal Monitoring remains recall-first but distinguishes confirmed freshness from temporal uncertainty. Relevant evidence without a reliable date is retained for human review. Retrieval time never substitutes for publication or event time. Known-source work is candidate-bound and capability-aware.

Figure 1. End-to-end Radar execution flow



2. Temporal Contract

Sources expose independent retrieved_at and published_at. Evidence exposes event_at, optional end date, temporal status, date basis, confidence and safe date evidence. Legacy observed_at is accepted only as retrieval time.

Temporal statuses are confirmed_in_window,

review_needed_date_unknown, review_needed_date_conflict and rejected_out_of_window. A fresh publication about a future plan can be a current signal. If no reliable publication/event date exists, relevant evidence is retained for review and never counted as a confirmed control.

Requirements: SM-TIME-01, SM-TIME-02, SM-TIME-03, SM-SCORE-01.

3. Source Capability And Candidate Binding

Every source is classified as identity_only, official_press, event_feed, project_or_asset_history, registry, generic_web or unknown. Every candidate/source relation is classified as matched_candidate, group_only, cross_entity, unknown_owner or no_url with basis, confidence and reason.

Only candidate-matched signal-capable URLs become known-source tasks. Identity/registry and excluded refs remain visible in a binding ledger. Unknown capability with matched ownership may be searched, but cannot confirm a signal without runtime capability and temporal validation. Rules are generic; company names and benchmark domains exist only in fixtures and validation docs.

Requirements: SM-CAP-01, SM-BIND-01, SM-BIND-02, SM-ARCH-02.

4. Planning And Recovery

The assembler preserves candidate aliases and candidate-specific source refs. Each candidate/criterion pair has at most two known-source tasks plus one official and one open-web task. Primary query uses the legal name and configured criterion evidence; one accepted alias query is available for bounded revision. Transport errors receive one primary retry before the lane becomes incomplete. Confirmed and unknown-review evidence receive stable source keys. Incremental runs project duplicate_existing_signal or duplicate_existing_review rather than republishing old rows.

Requirements: SM-QUERY-01, SM-RETRY-01, SM-DED-02, SM-AUD-02.

5. Expanded Benchmark

Post-run acceptance uses source candidate run radar-run-3bbf9c0f-330e-4468-8901-966a751234a8, six explicit candidates (three accepted and three review-needed), criteria S1 and S2, and exactly twelve candidate/criterion pairs. Control URLs, accepted equivalent URL sets and expected dates are loaded only after the run and must not occur in planning metadata.

The quality budget is 48 tasks, 60 provider calls, 8 extraction retries, 4 backup retries, 60 lookback queries, 120 source verifications and one revision per pair. Required acceptance lanes must not be budget-limited.

Requirements: SM-BENCH-01, SM-BENCH-02, SM-BENCH-03.

6. Validation And Closure

The validator matches controls by candidate, criterion, canonical URL or an explicit accepted URL set for the same public event, and expected date/range. It separately reports confirmed, unknown/conflicting, out-of-window, binding-rejected and duplicate evidence. Aggregate observed count is not a positive-control metric.

Two persisted Docker/API runs, an API restart and full requirement traceability are mandatory. This document becomes Implemented only after validation PASS; the implemented behavior is then reconciled into AS IS.

Requirement: SM-PROC-02.

7. Requirement Traceability

Mandatory IDs: SM-TIME-01, SM-TIME-02, SM-TIME-03, SM-CAP-01, SM-BIND-01, SM-BIND-02, SM-QUERY-01, SM-RETRY-01, SM-SCORE-01, SM-BENCH-01, SM-BENCH-02, SM-BENCH-03, SM-DED-02, SM-AUD-02, SM-ARCH-02, SM-PROC-02.

8. Implementation Evidence

The implemented validation pair is:

- first live quality run signal-run-010ef75d-c626-44e3-a025-56c95522c1a8;
- second incremental run signal-run-df00b3b8-267c-4091-a4dd-8167434e2cf3.

The first run produced six candidates, twelve candidate/criterion pairs, four observed outcomes, two unknown-date review items, zero retrieved-at freshness violations and zero confirmed out-of-window evidence. The post-run evaluator matched four of four positive controls, two negative controls and the required unknown-date control. The second run loaded previous source keys, used incremental windows and republished zero previous sources.